

组合起来

Compact CNN -- init -- (combine the different layers of
1. Layer : conv | conv2D(, conv1,2, pool1,2, relu1,2,
self : conv1 = conv2D(, Flatten, Linear)

in - channels = 3,

RGB

out - channels = 8,

kernel - size = 3,

conv kernel = size

stride = 1,

padding = 1

out-channels

input . shape (N, ^{in-channels} 3, 32, 32) $\xrightarrow{\text{conv1}}$ ~~At~~ (N, ^{out-channels} 8, 32, 32)

$$\text{Height} = \frac{(H + 2P - K)}{S} + 1$$

2. Layer ReLU 1

for all element from conv1 do $\max(x, 0)$

shape remains unchanged

3. Layer Pool 1

kernel size = 2

stride = 2

Let the space reduce half

for each channel do 2×2 max pooling, stride = 2

so: shape: $(N, 8, 32, 32) \rightarrow (N, 8, 16, 16)$

4. Layer conv 2

self.conv2 = Conv2D(
in_channels = 8,
out_channels = 16, def by me
kernel_size = 3
stride = 1
padding = 1,)

$(N, 8, 16, 16) \xrightarrow{\text{conv2}} (N, 16, 10, 10)$

5. Layer RELU

all element do $\max(x, 0)$ one by one.

shape remains unchanged.

6. Layer pool 2

self.pool2 = MaxPool2D(kernel_size=2,
stride=2)

$(N, 16, 16, 16) \rightarrow (N, 16, 8, 8)$

7. Layer flatten

$\uparrow 16 \times 8 \times 8 = 1024$

$(N, 16, 8, 8) \rightarrow (N, 1024)$

8. Layer classifier / Linear

in_features = $16 \times 8 \times 8$

out_features = 10

$\rightarrow$ why 10?

$(N, 1024) \rightarrow (N, 10)$ because CIFAR-10 has 10 types.

the final output of compact CNN forward is logits.

it isn't probabilities.

(logits is ~~is~~ scores from the model for 10 types,
(def in lecture))

e.g.: a photo ---

1.2, -0.5, 0.3, 2.1, -1.0, 0.0, 0.0, -0.2, 1.5, 0.9

softmax classifier output $\rightarrow$ Loss